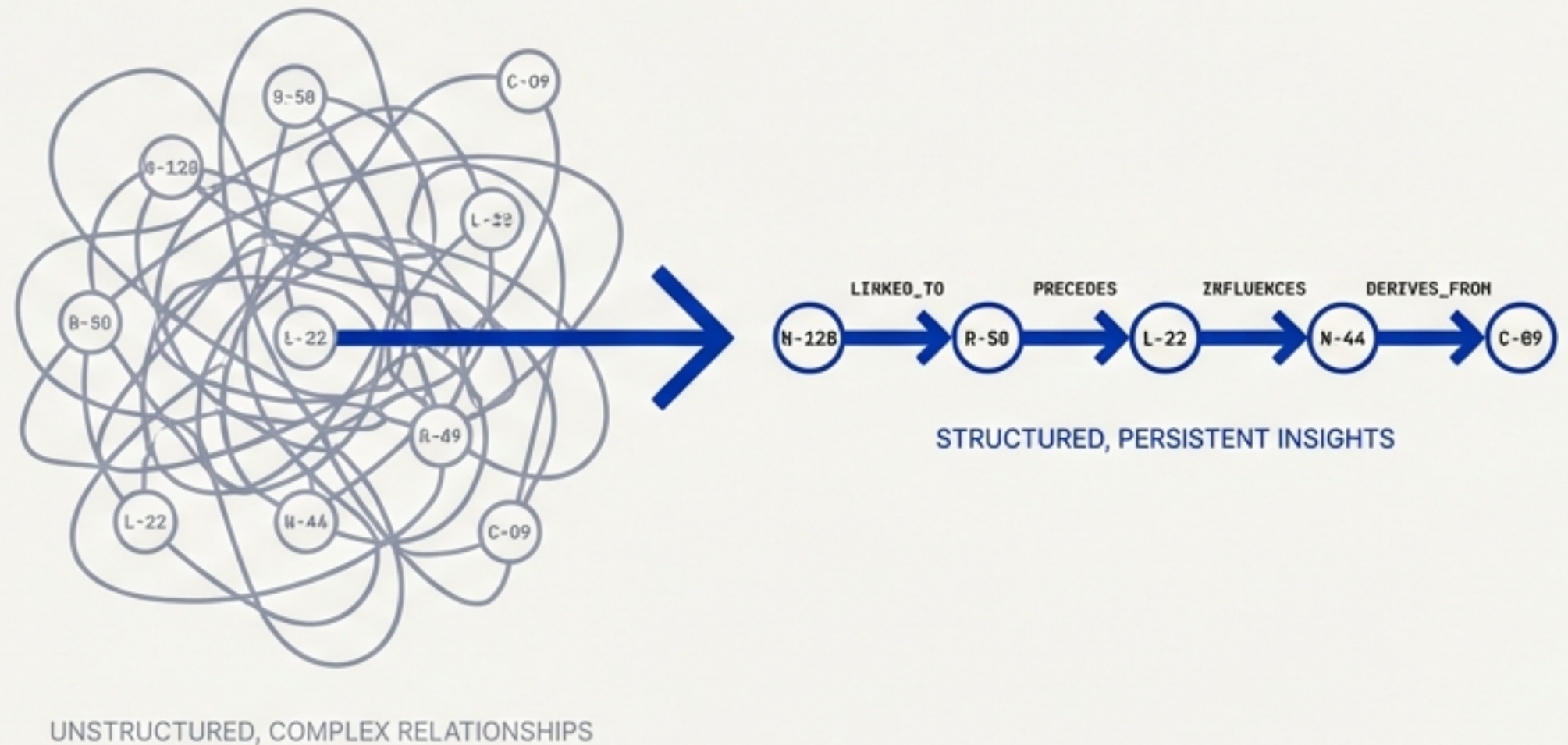


Leveraging Static Graph Visualisations for Collaborative Knowledge Management

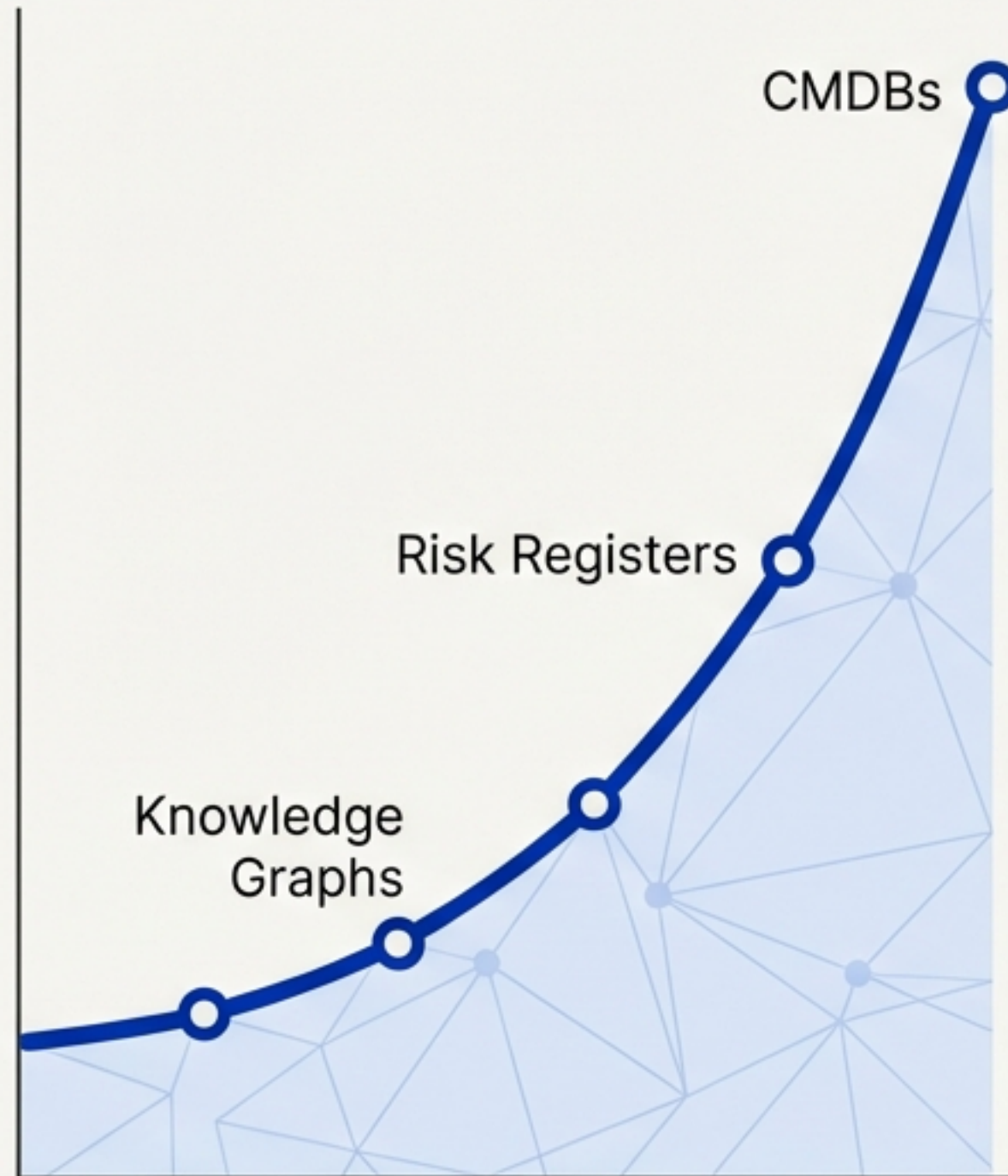
Turning complex relationships into persistent, actionable insights.



Based on the whitepaper "Leveraging Static Graph Visualizations for Collaborative Knowledge Management"

The Data Paradox: More Graphs, Less Shared Understanding

Data Generation



Actionable Insight

The “So What?” Dilemma



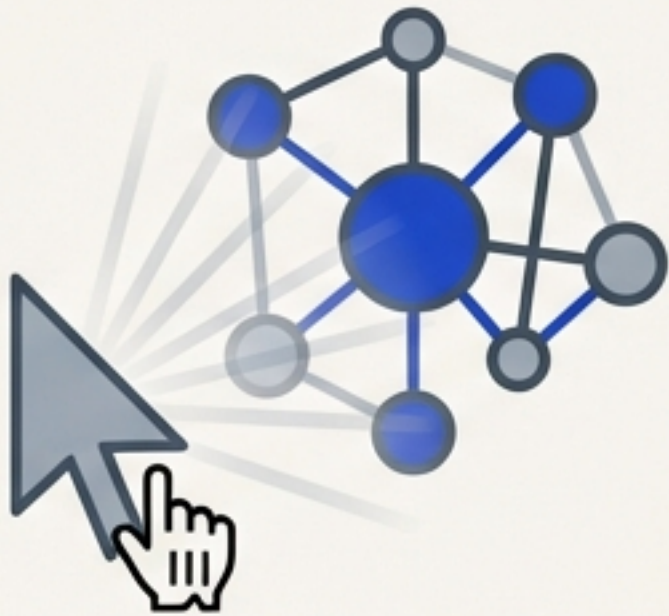
Organisations are increasingly adopting graph-based data structures. However, teams struggle to extract actionable insights from these elaborate graphs.

The Gap: While LLMs and automation assist in *populating* graph data, the ability to *communicate* that data has not kept pace.

Key Insight: Data ownership is meaningless without the ability to share a consistent snapshot with decision-makers.

Why Traditional Visualisation Fails Collaboration

Interactive tools are designed for solitary exploration, not team communication.



Ephemeral Views

Insights gained by panning and zooming exist only in the moment. You cannot 'version control' a mouse hover.



The Hairball Effect

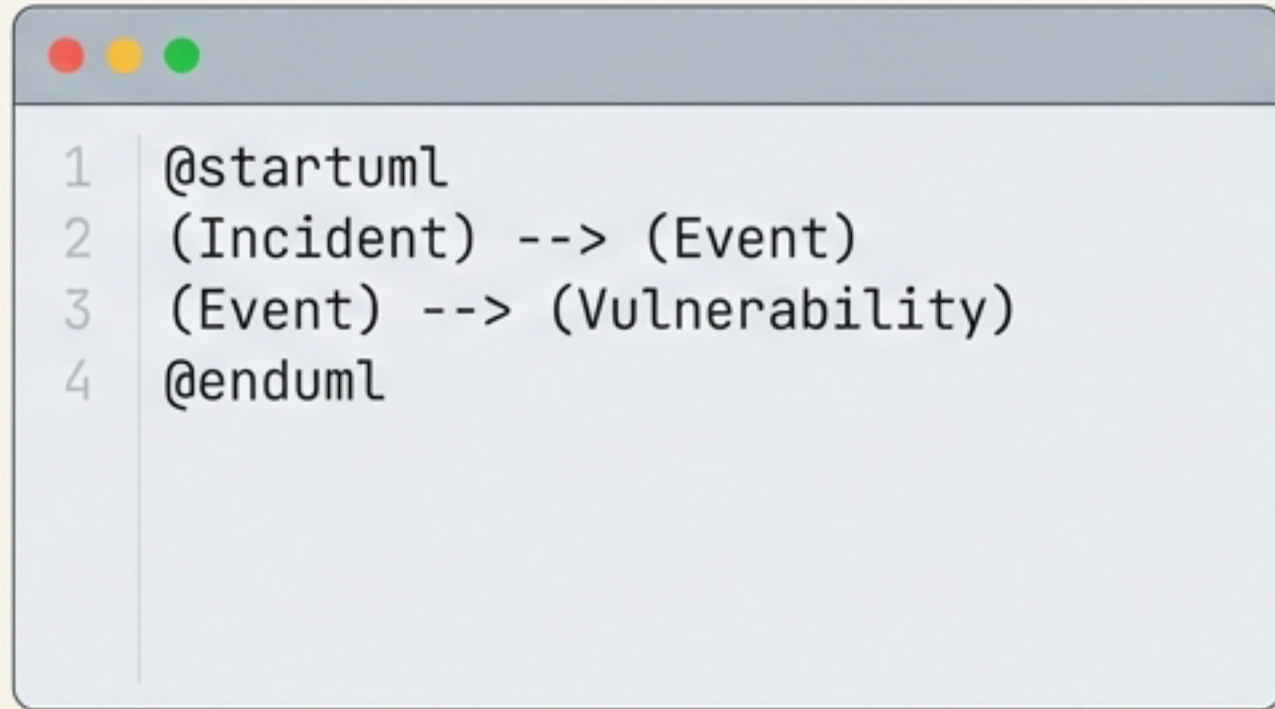
Naively visualising all relationships creates an unreadable blob where critical details are obscured by density.



The Silo

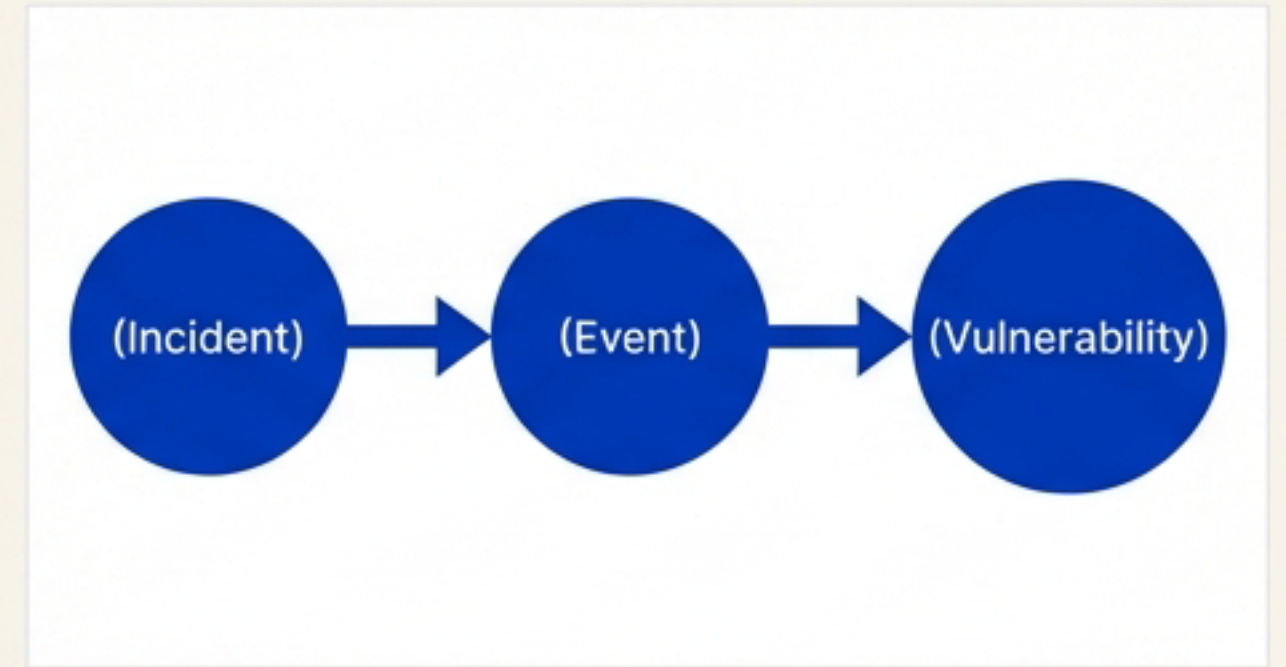
Insights remain locked in specialised tools like Neo4j interfaces that non-technical stakeholders never access.

The Solution: Static, Regenerable Snapshots



Diagrams as Code

Auto-Render

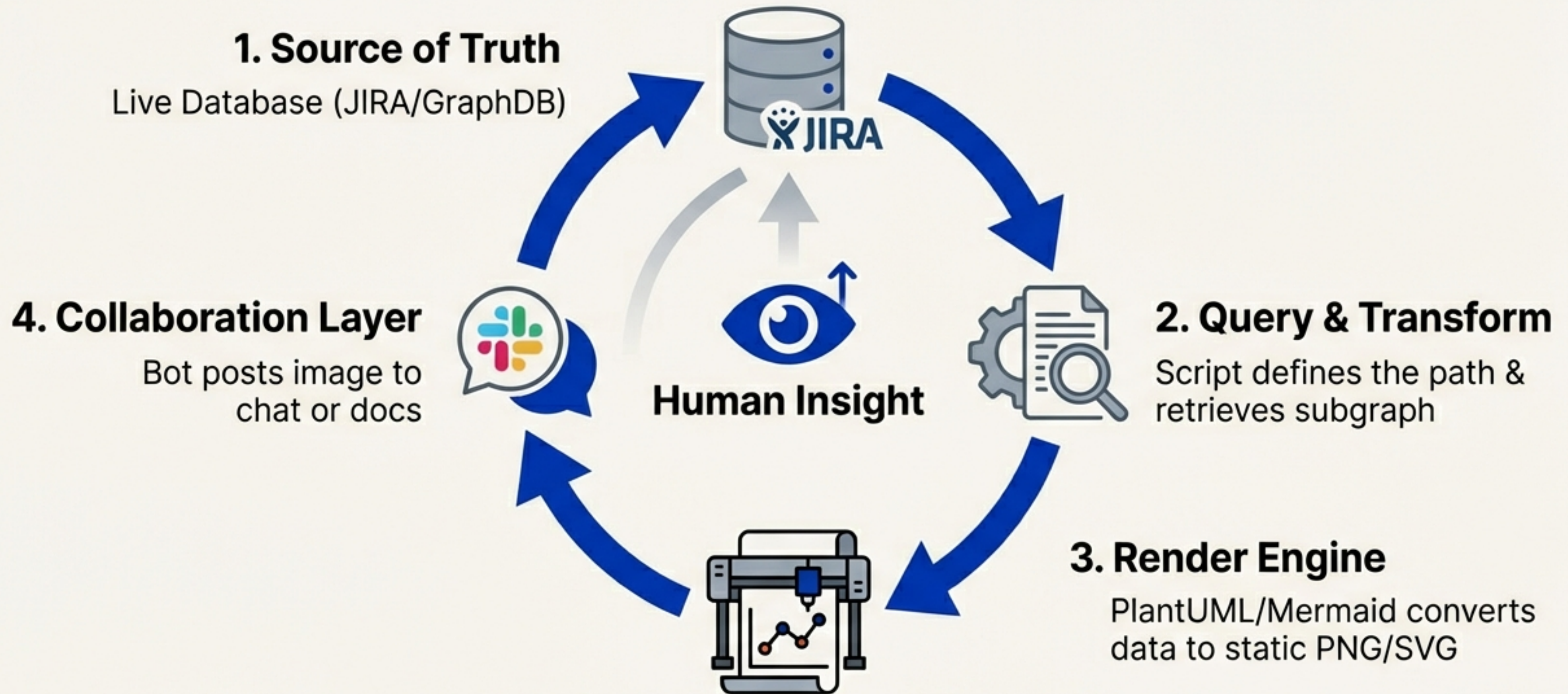


Reproducible Artifact

Core Concept: Treat graph visualisations as ephemeral artifacts generated from a single source of truth.

1. **Continuous Generation:** Snapshots are updated automatically when data changes.
2. **Reproducibility:** A “test script” for organisational knowledge.
3. **In-Context Delivery:** Images pushed directly to Slack, Teams, or Jira.

The Workflow Architecture

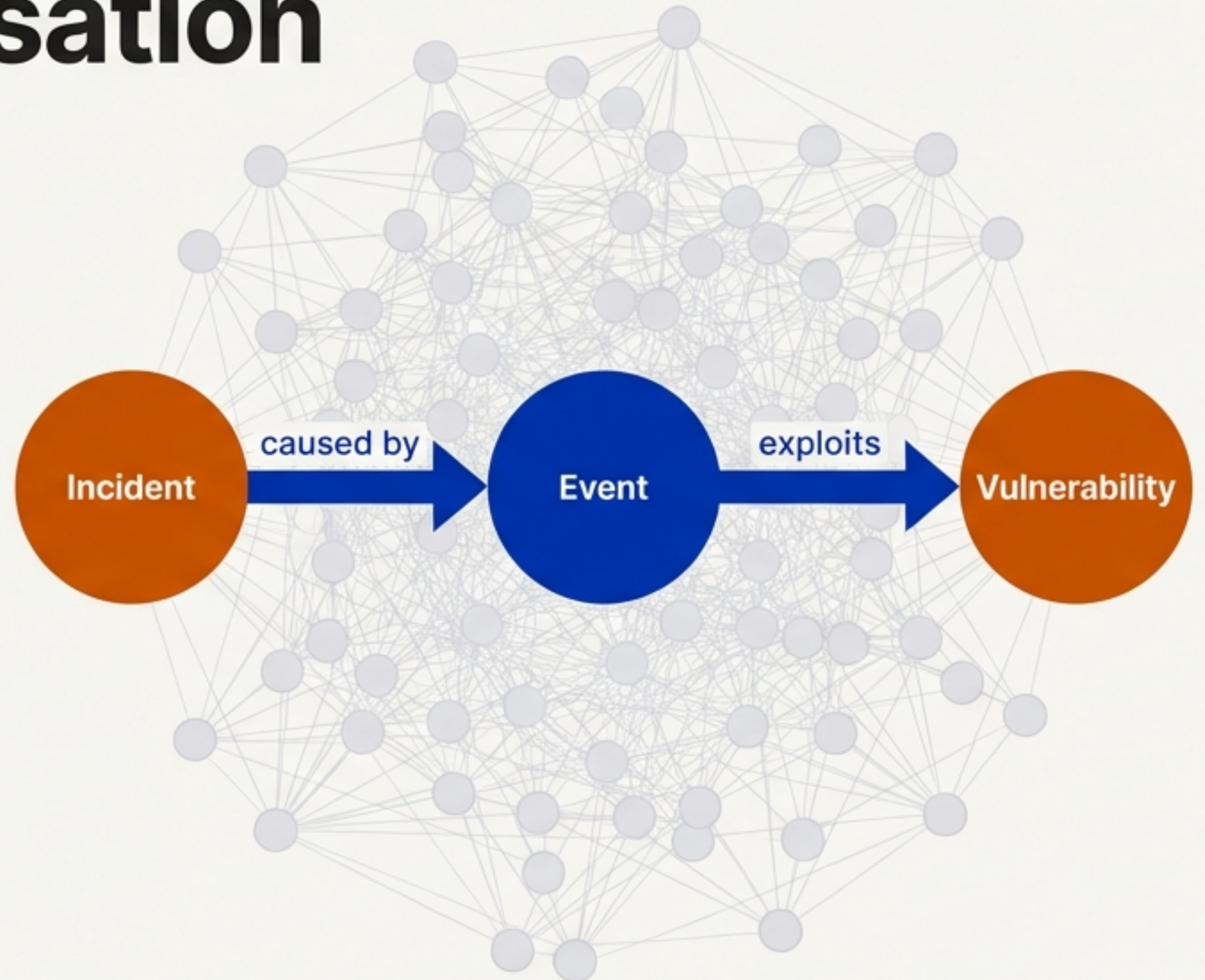


Taming the Hairball: Path-Based Visualisation

Core Strategy: Do not show the whole graph. Show the relevant thread.

Inter Regular

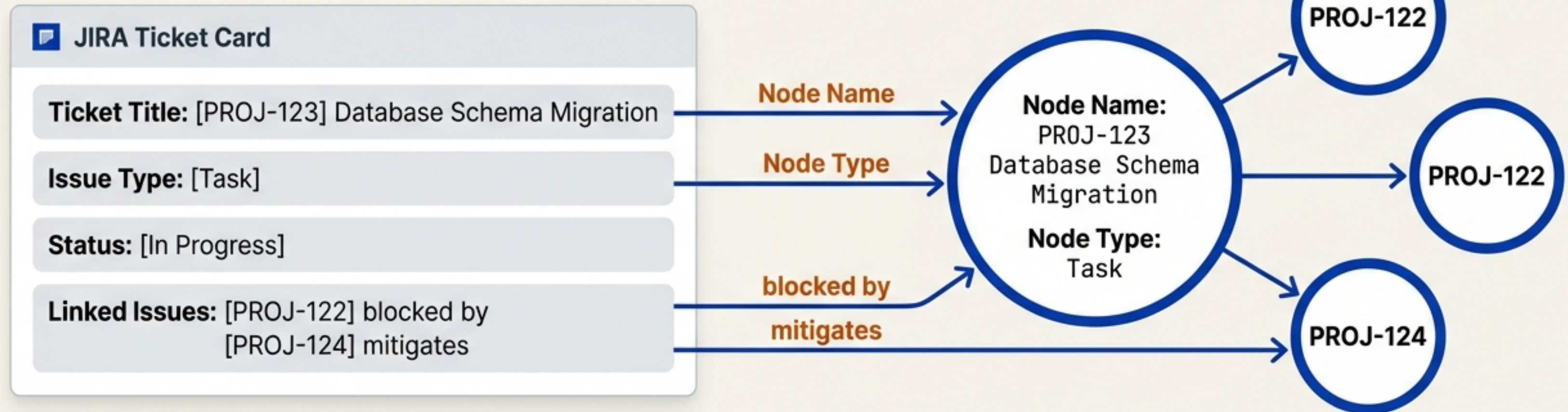
- Traverse, Don't Expand: Follow a semantic path driven by a specific question.
- The 'Verb' Approach: Traverse specific link types to build a narrative.
- Result: A linear tree that tells a clear story, eliminating visual noise.



The Unlikely Hero: JIRA as a Graph Database

Insight: You likely do not need new infrastructure; you need better usage of existing tools.

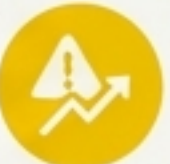
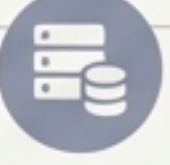
- **Graph Structure:** Issues = Nodes, Links = Edges.
- **Semantic Rigour:** Define strict link semantics (e.g., “blocked by”, “mitigates”) to turn JIRA into a relationship engine.
- **Accessibility:** Non-technical staff edit the graph using familiar forms.



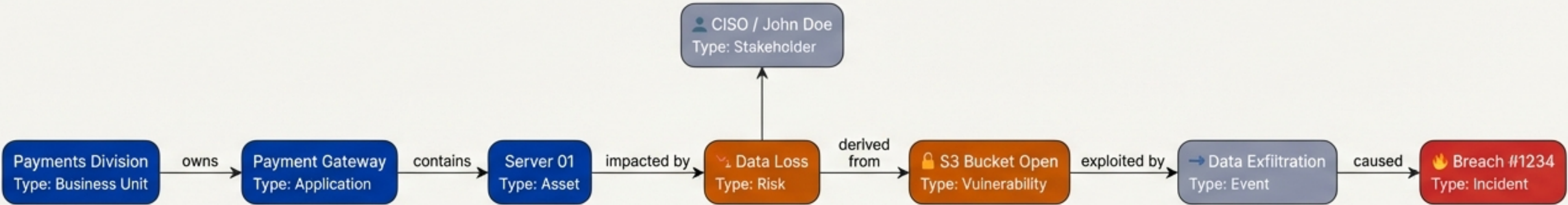
Case Study: Graph-Based Risk Management

Context: An enterprise security team managing Incidents, Vulnerabilities, and Risks.

The Goal: When an incident occurs, immediately visualise its full context—root cause upstream and business impact downstream.

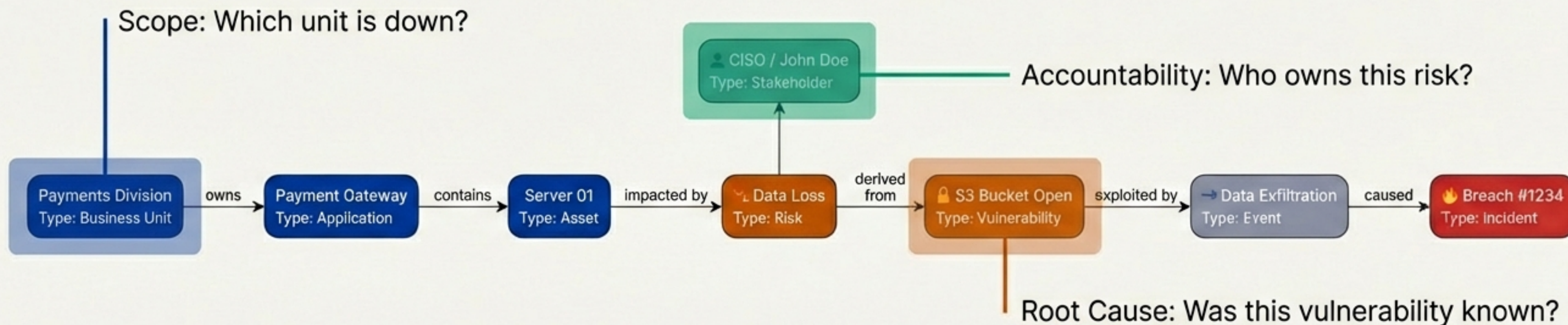
Node Types		Edge Types	
	Incident		exploits → 
	Event		causes → 
	Vulnerability		has risk → 
	Risk		owns → 
	Stakeholder		
	Asset		impacts → 

The ‘Incident Context’ Visualisation



Automatically Generated PlantUML Output

From Image to Action



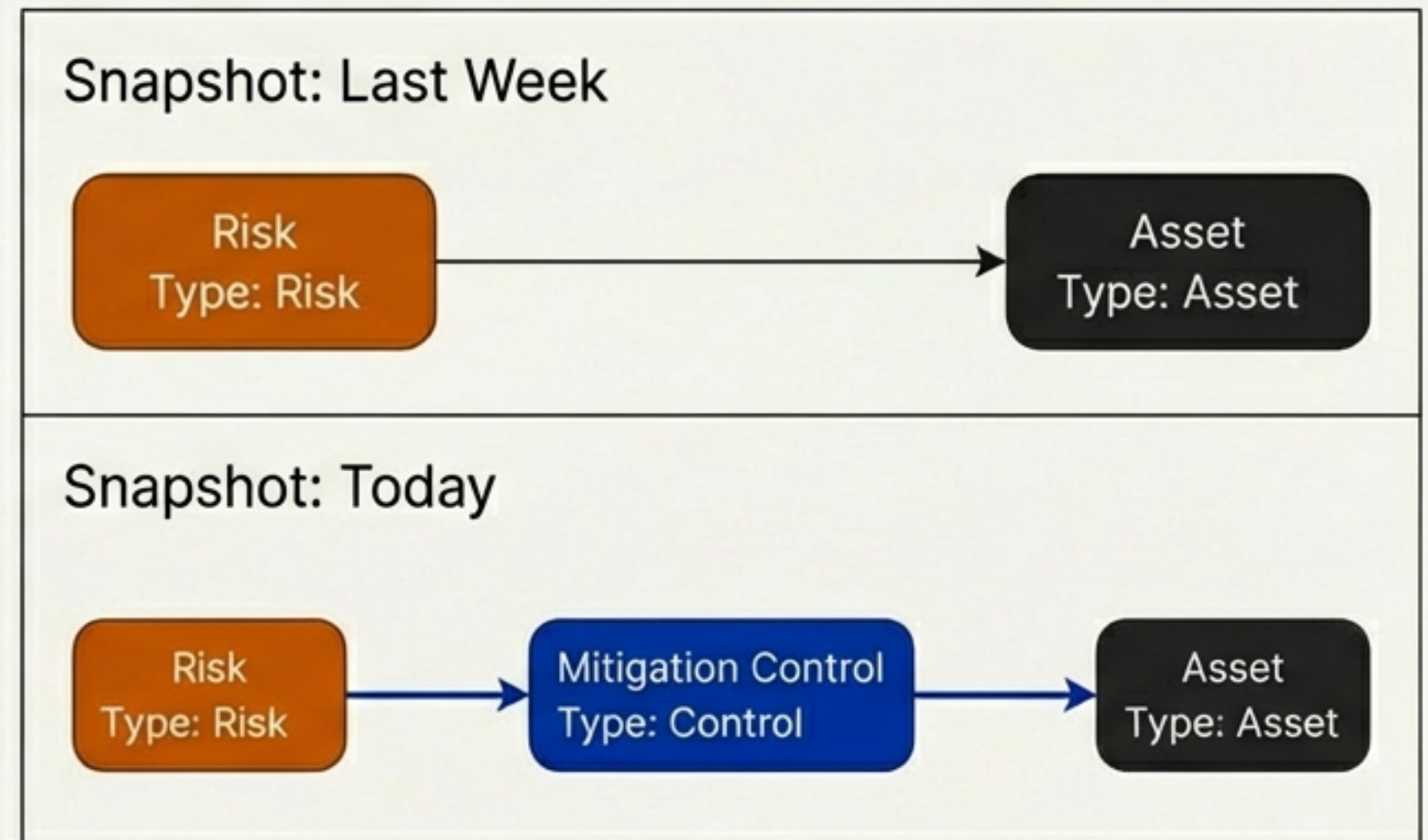
The static image answers critical questions instantly without query languages.
Outcome: Fact-based discussions occur in Slack, driven by a shared, indisputable image.

Strategic Benefit: Persistent Visual History

The 'Visual Diff':

- By archiving snapshots, you create an audit trail.
- Compare 'The Network Last Week' vs. 'The Network Today.'
- Essential for post-mortems and proving what was known at a specific point in time.

Before and After



Strategic Benefit: Democratised Insights

Bringing the graph to the people, not the people to the graph.

Slack

#security-incidents

New Insight Shared: Payment Flow Analysis

The static image answers critical questions instantly. 10:45 AM

+ @ 🎤

Subject: Weekly Risk Report - Payments Division

Team, attached is the latest visual analysis to be the graph to diiison - Payments Division.

Fact-based discussions occur here, driven by this image.

Reply Forward

MONTHLY SECURITY OVERVIEW Page 5

Section 3: Critical Vulnerability Analysis

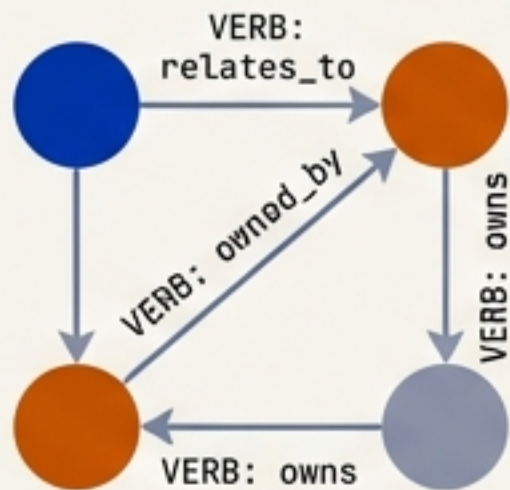
The image below details the recent breach pathway, highlighting the open S3 bucket as the root cause. Errors in the image prompt immediate data correction in the source.

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- **Zero Learning Curve:** Executives don't need to learn Cypher or Gremlin.
- **Workflow Integration:** Users interact with images in tools they already use.
- **Rapid Feedback Loop:** Errors in the image prompt immediate data correction in the source.

Implementation Guide

1. Data Modeling



Define clear Node types and Link verbs (ensure inverses exist).

2. Sync Strategy



Sync API data to a query-friendly store (e.g., ELK stack) for speed.

3. Tooling Selection



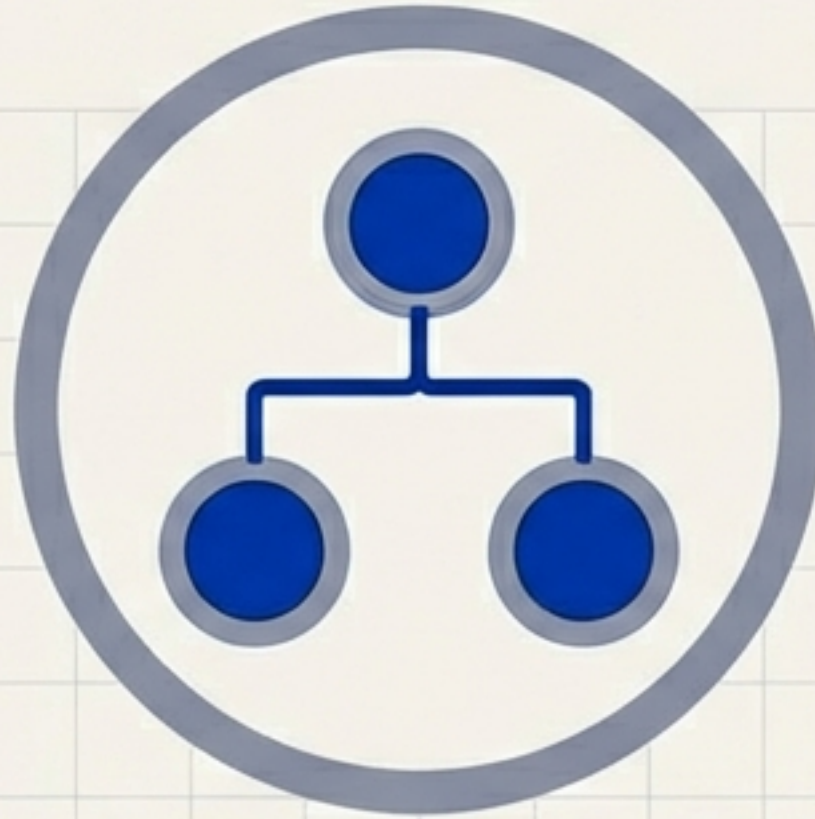
Renderer: PlantUML (Structure) or Mermaid (Docs). Automation: Serverless/Bots.

4. Aesthetics



Use consistent colour coding to reduce cognitive load.

Conclusion: Build Graphs for Collaborators



Stop building graphs for explorers; start building them for collaborators.
Final Takeaway: By automating the creation of focused, static snapshots, we turn ephemeral data into enduring organisational knowledge.

Next Step: Identify one complex process (Risk, Dependency, Org Chart) and build a snapshot pipeline for it today.